

School of Arts and Sciences

Department of Computer Science and Information Technology

Fall 2025 – 2026

CSCI410 Mobile Application

Class Day : _____ **Class Time:** _____ **Room:** _____
Instructor : _____
E-mail : _____
Office : _____ **Office Hours:** _____ **Extension No:** _____

I. Course Description

This course teaches students how to build, test and deploy mobile applications. Students will learn how to take their innovative ideas from conception to the market through a series of Labs and group project. This is an introductory course aimed at undergraduate students who have programming experience. However, there is a significant amount of programming requiring a commitment on the part of the student.

II. Course Objectives

CO-1: Build, test and deploy mobile applications

CO-2: Take their innovative ideas from conception to the android market through a series of labs and projects.

III. Course Learning Outcomes

CLO-1: Describe the major components needed for mobile application development.

CLO-2: Test mobile applications using emulators and deploying them to real life devices.

CLO-3: Create responsive GUI (Graphical User Interface) for mobile applications.

CLO-4: Create REST services and link them to online databases.

CLO-5: Create frontend mobile applications to access REST services.

IV. Mapping to program outcomes

Mobile Application	PLO-1	PLO-2	PLO-3	PLO-4	PLO-5	PLO-6	PLO-7	PLO-8	PLO-9
CLO-1	X								
CLO-2						X			
CLO-3				X					
CLO-4				X	X			X	X
CLO-5				X	X			X	X

V. Topics Covered

Configuring a mobile applications' development environment

Testing and deploying your applications

Using the different GUI controls and responsive layouts

Configuring an online environment for REST services and online databases

Understanding the concepts of frontend applications

Using a framework to create frontend applications

VI. Prerequisite

CSCI300 Intermediate Programming with Objects

CSCI335 Database Systems

VII. Course Material:

Flutter for Beginners: An introductory guide to building cross-platform mobile applications with Flutter and Dart 2, **Publisher:** Packt Publishing; **ISBN-10:** 1788996089

VIII. Software required(for computer-lab courses)

Android Studio, Flutter SDK and Git

IX. Grading Policy and Withdrawal Deadline

Project Part I	15% Week 09
Project Part II	15% Week 14
Midterm Exam	35% Week 10
Final Exam	35%

Withdrawal Deadline **Week 14**

I. Detailed Schedule

Week		Book Reference	Topic Details
Week 1 (Sep 29 – Oct 2)	Lecture 1	Flutter for Beginners	Syllabus: reading and explaining. Chapter 1: Syllabus: reading and explaining. Software needed for flutter development Introduction to Dart language Dartpad Writing a simple dart hello world Simple program to add 2 integer numbers
	Lecture 2	Flutter for Beginners	Chapter 1 Dart operators Dart types and variables Control flows and looping
Week 2 (Oct 5 - 9)	Lecture 1	Flutter for Beginners	Chapter 1: Functions Lists
	Lecture 2	Flutter for Beginners	Chapter 1: Maps, Sets, and Packages
Week 3 (Oct 13 - 16)	Lecture 1	Flutter for Beginners	Chapter 2: Dart classes setters/getters toString method constructors
	Lecture 2	Flutter for Beginners	Chapter 2: Class inheritance Abstract classes <u>Interfaces</u>
Week 4 (Oct 20 - 23)	Lecture 1	Flutter for Beginners	Chapter 3: Hello Flutter Stateless widgets Stateful widgets Creating a Flutter Project Creating an Emulator Hello World Application MyApp StatelessWidget Displaying text using the Text Widget Adding the Scaffold Class Adding an AppBar
	Lecture 2	Flutter for Beginners	Chapter 4: Creating a custom widget class Adding a Column widget Adding a Row widget

Week 5 (Oct 27 - 30)	Lecture 1	Flutter for Beginners	Chapter 4: Interactive applications The StatefulWidget Class The TextField Widget Creating a custom TextField Widget
	Lecture 2	Flutter for Beginners	Chapter 4: The ElevatedButton Widget Adding Images
Week 6 (Nov 3 - 6)	Lecture 1	Flutter for Beginners	Chapter 4: Selection widgets DropDownMenu Widgets
	Lecture 2	Flutter for Beginners	Chapter 4: Checkbox Radio button
Week 7 (Nov 10 - 13)	Lecture 1	Flutter for Beginners	Chapter 7: Flutter Routes The Navigator Class The Navigator Push Function
	Lecture 2	Flutter for Beginners	Chapter 7: Flutter Icons The TextEditingController Adding a dispose destructor
Week 8 (Nov 17 - 20)	Lecture 1	Instructor notes	Create a ListView Add items at runtime to a ListView
	Lecture 2	Instructor notes	Customize ListView items Display Images from a remote server
Week 9 (Nov 24 - 27)	Lecture 1	Instructor notes	Solve exercises
	Lecture 2	Instructor notes	Project 1 presentations
Week 10 (Dec 1 - 4)	Lecture 1	Instructor notes	Create a cloud account Create an online database
	Lecture 2	Instructor notes cs	Midterm Chapter 1, 2, 3, 4, 5 and ListViews
Week 11 (Dec 8 – 11)	Lecture 1	Instructor notes	Create an application to retrieve data from and REST service
	Lecture 2	Instructor notes	Create an application to add data to online database
Week 12 (Dec 15 - 18)	Lecture 1	Instructor notes	Create an application to delete data from online database
	Lecture 2	Instructor notes	Updating online databases values

Week 13 (Dec 22 - 25)	Lecture 1	Instructor notes	<i>December 24 and 25 - Christmas Vacation</i> Solve exercises
Week 14 (Jan 5 - 8)	Lecture 1	Instructor notes	<i>January 6 – Armenian Christmas</i> Project 2 presentations and general review
	Lecture 2	Instructor notes	Project 2 presentations and general review

I. Class Rules & Discipline

Attendance Policy

The student is administratively withdrawn (AW) from the class if he/she is absent for more than one third of the classes (15 class sessions for courses offered 3 times a week, 10 for courses offered twice a week, and 5 for courses offered once a week). If the number of absences exceeds the limit after the withdrawal deadline, i.e. including all the period up-to the end of the semester, the student will not receive an AW grade, therefore, the final examination must be written, otherwise, an F grade will be granted. When absent, you are responsible for all material presented in class. Missed exams will only be excused by documented evidence as explained in the make-up examination policy below.

A “10%” has been allocated on the “Marking Profile” for attendance. A student who hasn’t missed any lecture will get a full mark on the attendance grade. Every absence, a 20% will be deducted, that is, a student with 5 absences will get a ZERO on that grade.

Cheating

The University policy on cheating will be strictly applied. Students are required to adhere to the policy on academic irregularities contained in the university student handbook. Collaboration among students in solving assignments is not acceptable. It is okay to ask another student for advice when stuck on a problem, but if the instructor receives assignments that are substantially equivalent, all involved students will be punished with a grade of zero.

Classroom Manners

All students are expected to conduct themselves in a professional manner. Students should not: keep their cell phones on during the class, disrupt the class, do other course work in class, have side conversations with other students, be rude towards anyone. The first failure to respect these rules will result in a *warning* from the instructor. If a student chooses to disregard the *warning*, he/she will be dismissed from class.

Instructor late to the class

In case the instructor is late the students should wait for 15 minutes before leaving the class.

Abuse of Computing Privileges

The Office of Information Technology maintains policies posted in the computer labs regarding proper behavior in the labs. Failure to adhere to these policies may result in loss of computer privileges, and possible legal action.

Make-Up Examination Policy

As per university council decision, a student is eligible for a mid-term or final examination make-up if and only if he/she had the following incidents:

- a. Sickness; **proved by hospitalization report**; that is; a discharge summary is necessary.
- b. Death in the family **proved by a death certificate** and personal identification.
- c. Accidents **proved by an expert report**.

A student that misses an exam should submit a petition within one week from the examination date. Late petitions will be rejected unless it is an extreme case (Hospitalization as an example).

In case a petition is accepted, the student submits a MAKE-UP exam. The grade of the missing exam will be equal to a percentage % of the Final Exam's grade. The percentage % will be set by the administration once the petition is approved.

Cheating policy: Any cheating will be rewarded with a zero, all students involved in cheating will get a zero (even if the cheating is in one question). The activity would not be repeated.

Missing Assignments and Projects Policy

- A -10 per day penalty will be applied to late projects (weekends count as two days).
- A ZERO will be applied for late assignments on the next day.

Curriculum, Yearly Plan, and Advising

If a student majoring in Computer Science or Information Technology is planning to graduate in three years then he/she should follow the corresponding major yearly plan. A copy of the yearly plan and the curriculum (which contains the prerequisites and co-requisites of courses) can be picked up from the office of the school of Arts and Science. When not sure about anything please contact your advisor prior to or during the registration period.

Probation Policy

If a student who has attempted 19 credits has a GPA or a major GPA below 2.0 then the student is considered under probation.

E-mails

Students should use their university e-mails in communicating with the instructors. Important announcements, syllabi, assignments, and solutions will be sent to your university e-mail accounts.

University Peer Review Policy

All classes will be visited by instructors either from the same or different departments. This policy helps instructors figure out ways to improve student educational experiences in class and eventually their performance.

Snapshots

Students are not allowed to take snapshots of the whiteboard in classrooms using their mobile-phones.

Equivalent Lettering Grades

<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>	<u>Percent (%)</u>	<u>QPTS</u>	<u>Letter Grade (G)</u>
<60	0	F	67	1.7	C
60	1	D	68	1.8	
61	1.1		69	1.9	
62	1.2		70	2	
63	1.3		71	2.1	
64	1.4		72	2.2	
65	1.5	D+	73	2.3	B+
66	1.6		74	2.4	
75	2.5	C+	83	3.3	
76	2.6		84	3.4	
77	2.7		85	3.5	
78	2.8		86	3.6	
79	2.9		87	3.7	
80	3	B	88	3.8	A
81	3.1		89	3.9	
82	3.2		>=90	4	

<u>Grade</u>	<u>Description</u>	<u>QPTS</u>
A	Excellent	4.00
B+	Very Good	3.50
B	Good	3.00
C+	Above Average	2.50
C	Average	2.00
D+	Fair	1.50
D	Poor	1.00
F	Fail	0.00
IP	Course in Progress	
I	Incomplete work	
W	Student has withdrawn from the course	
AW	Administrative withdrawal from the course	
R	Course has been repeated	
S	Satisfactory work	
U	Unsatisfactory work	
%G	Final Grade	
QPTS	Quality Points	

Computer Science Outcomes

Graduates of the Computer Science Program will have the ability to:

	Program Outcomes
category -1 Underlying Conceptual Basis for Informatics	PLO -1 Describe and explain the essential facts, concepts, theories and mathematical methods relevant to computing, computing equipment, computer communication and informatics applications as appropriate to their program of study
	<i>PLO-1 for CSIT=> Gain basic skills in mathematics and outline the characteristics of relevant state-of-the-art hardware and software and their practical application</i>
category -2 Analysis	PLO -2 Select and use relevant analytic, modelling and simulation methods
	PLO-3 Choose appropriate solution patterns, algorithms and data structures
category -3 Design and Implementation	PLO -4 Describe the software development process and apply relevant practical and programming skills to the creation of computer programs, and/or other informatics artefacts
	<i>PLO-4 for CSIT=>Specify and design computing/network hardware/software which meet specified requirements</i>
category -4 Economic, legal, social, ethical and environmental context	PLO -5 Explain the importance of information privacy and security issues in relation to the design, development, maintenance, monitoring and use of informatics-based systems
category -5 Informatics practice	PLO -6 Describe and explain management techniques appropriate to the design, implementation, testing, deployment and maintenance of informatics systems, including project management, configuration management, change management, etc., and including relevant automated techniques
	<i>PLO -6for CSIT=>Demonstrate an awareness of project management and business practices, such as risk and change management, and understand their limitations;</i>
category -6 Other Professional Skills and Competences	PLO -7 Communicate effectively both verbally and using a variety of communications media
	PLO -8 Plan self-learning and improve personal performance as a foundation for lifelong learning and ongoing professional development
	PLO -9 Participate effectively in informatics group-working